Speaker Script · Numbers That Talk: divisibility rules and the modular arithmetic behind them

Slide 1 · Title: Numbers That Talk (Zhiheng, about 30 s)

Hi everyone. I'm Zhiheng, and this is Yueran. You should all have a worksheet. Quick question to start. This year is 2026. **Is 2026 divisible by 9?** Add the digits: 2, 0, 2, 6. That's 10. Not a multiple of 9, so no. Here's the weird part. 10 leaves remainder 1, and so does 2026. Why? Let's find out with a trick.

Hand out the worksheets before you start. After the question, wait 5 seconds so people can fill in the warm-up line.

Slide 2 · Magic trick 1: the vanishing digit (Zhiheng, about 90 s)

Look at box 1 on your worksheet. Write down a four-digit number. Just don't use the same digit four times. Now shuffle the digits and write the new number. Subtract the smaller one from the bigger one. Done? Good. **Circle one digit in your answer. Not a zero.** Keep it secret. Can someone read me the other digits? Any order. ... Your circled digit is [answer]. Right? One more person. ... OK. I'm not reading your mind. The number 9 is doing the work.

Give people about 20 seconds to write. Don't rush them.

Add the digits they read in your head. Answer = what is missing to reach the next multiple of 9. If the sum is already a multiple of 9, the answer is 9.

Backup: 8317 → 1378 → 6939, hide a 9, hear 6, 3, 9, sum 18, answer 9. If someone gets 0, they shuffled into the same number. Ask them to shuffle again.

Slide 3 · A clock with only 9 marks (Zhiheng, about 90 s)

Here's how it works. 2026 is 2 thousands, 0 hundreds, 2 tens, and 6. **Now look at this clock. It only has 9 marks.** Start at the top and walk 10 steps. You land one mark past the top. Same as walking 1 step. Walk 100 steps? Still one mark. 1000? Still one. So on this clock, 2026 is just 2 plus 0 plus 2 plus 6. In class words: 10 is congruent to 1 mod 9. That's the whole idea. Now the trick. Your number and the shuffled one use the same digits. So they stop at the same mark. Subtract, and you're back at the top. That's zero. **So the digits of your answer always add up to a multiple of 9.** You told me all but one. I just filled the gap. That's also why you can't circle a zero. A zero and a nine look the same to me.

Walk your finger 10 steps around the clock, slowly.

Slide 4 · Same idea, bigger chunks (Zhiheng, about 45 s)

Same idea, bigger steps. 1000 is 999 plus 1. So on a 999-clock, 1000 acts like 1. And 999 is 27 times 37. So cut a number into three-digit chunks from the right and add them. **45658. Chunks 45 and 658. Add them: 703. And 703 is 19 times 37.** So 37 divides 45658. No long division. Just remember to cut from the right.

Slide 5 · What about 7? (Zhiheng, about 30 s)

So far every rule comes from one thing: a power of 10 acting like 1. **But 999 is 27 times 37. No 7. No 13.** So does 7 have a rule at all? I thought no. Turns out there's a second family, where a power of 10 acts like minus 1. Yueran?

Hand over. Yueran asks everyone to open the calculator on their phone.

Slide 6 · Magic trick 2: 1001 brings it back (Yueran, about 90 s)

Thanks. Box 2 on your worksheet. Write a three-digit number. Now write it again right after, so you get six digits. Like 572572. Grab a calculator, your phone is fine. **Divide by 7.** It works, trust me. **Now divide that by 11. And divide by 13.** What did you get? Your number came back. Every time. Can someone tell me theirs? ... So why does this happen?

Give people about 30 seconds on their calculators.

572 → 572572 → ÷7 = 81796 → ÷11 = 7436 → ÷13 = 572. Works with a zero: 205 → 205205 → 29315 → 2665 → 205. A decimal means a typo.

Slide 7 · Why it works: 1000 acts like –1 (Yueran, about 60 s)

Writing abc twice is abc times 1000, plus abc. That's abc times 1001. And 1001 is 7 times 11 times 13. So dividing by 7, 11 and 13 is just dividing by 1001. That's the trick. Here's the useful part. 1000 is 1001 minus 1. So mod 1001, 1000 acts like minus 1. Cut a number into three-digit chunks and alternate plus and minus. You get the remainder mod 7, 11 and 13, all at once. **Try 2026. Chunks 2 and 026. 26 minus 2 is 24.** 24 mod 7 is 3. 24 mod 13 is 11. Both right.

Slide 8 · One idea, every prime (Yueran, about 90 s)

So here's the big picture. **Find a k so that 10 to the k is 1 or minus 1 mod d.** If it's 1, add k-digit chunks. If it's minus 1, alternate them. 9 uses k equals 1. 37 uses 3. 7 and 13 use 3, with the minus. Does every number have a k? **Fermat's little theorem says yes, for every prime except 2 and 5: 10 to the p minus 1 is 1 mod p.** The shortest k is called the order of 10 mod p. And here's what surprised us most. That same number is how long 1 over p repeats. **1 over 7 is 0.142857, repeating. Six digits. The order of 10 mod 7 is also 6.** Look at the remainders when you divide 1 by 7: 1, 3, 2, 6, 4, 5, then back to 1. Those are the powers of 10 mod 7. Six steps, six digits. Bonus: 142857 times 2 is 285714. Same digits, just rotated.

Trace the remainder cycle with your finger: 1 → 3 → 2 → 6 → 4 → 5.

Slide 9 · One sentence (Yueran, about 30 s)

One sentence. **If you know what 10 is mod d, you can build the rule for d yourself.** The proofs are on the back of your worksheet. More stuff, like a quick rule for 7 and why ISBN numbers use 11, is on our GitHub. Thanks! Go try the 1001 trick on a friend.

Likely questions (one or two sentences each)

Q1. How is this different from the talk on the rules for 3, 7, 9 and 11?

We only use those rules as a warm-up. Our question is why every number coprime to 10 has a rule, how short it can be, and why that length is exactly the period of 1 over p.

Q2. Why don't 2 and 5 have a digit-sum rule?

Because 10 is 0 mod 2 and mod 5. 10 is not a unit there, so it has no order. But 10 being 0 is exactly why the last digit, or the last k digits, is enough.

Q3. Does every number have a rule?

Every d coprime to 10 does; Euler gives 10 to the phi of d is 1. A general d splits into a power of 2, a power of 5 and a part coprime to 10. Test each part on its own.

Q4. Is there a shorter rule for 7?

Yes. Chop the last digit y and compute x minus 2y, repeat. 2026 goes to 202 minus 12, that is 190, then 19, not a multiple of 7. It uses the inverse of 10 mod 7, which is 5, or minus 2. It only keeps yes or no, not the remainder.

Q5. Can I start the alternating sum from the left?

For yes or no, yes. For the remainder, no: with an even number of digits the sign flips. 2026 from the left gives minus 2, which is 9, but the remainder is 2.

Q6. Why does 1/7 repeat every 6 digits but 1/37 every 3?

The period of 1 over d is the order of 10 mod d. The order mod 7 is 6; mod 37 it is 3, because 999 is 27 times 37.

Q7. Can the tricks fail?

Trick A fails only if someone circles a 0, since 0 and 9 look the same, or shuffles into the same number. Trick B cannot fail.

Q8. How is this related to cryptography?

The order of an element and primitive roots are the core of Diffie-Hellman and ElGamal. Check digits, like ISBN-10 with mod 11, are divisibility rules used in practice.

Q9. Why chunk from the right?

Because n is B zero plus B one times 10 to the k plus B two times 10 to the 2k and so on. B zero has to be the lowest k digits.

• 99 = 9 × 11；999 = 27 × 37；1001 = 7 × 11 × 13；99999 = 9 × 41 × 271。